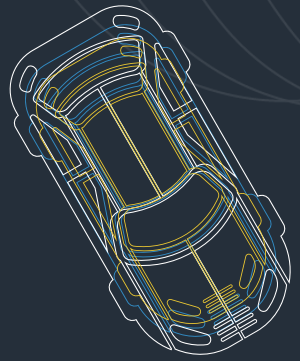
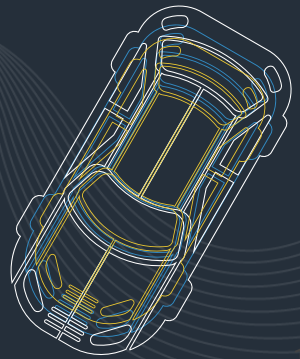




# Hydrogen Vehicle Revitalization

**Justin Chai and Ben Horvat**
**Advisor: Prof. Everbach**
Swarthmore College E90
Spring 2024



# That Thing In the Highbay?







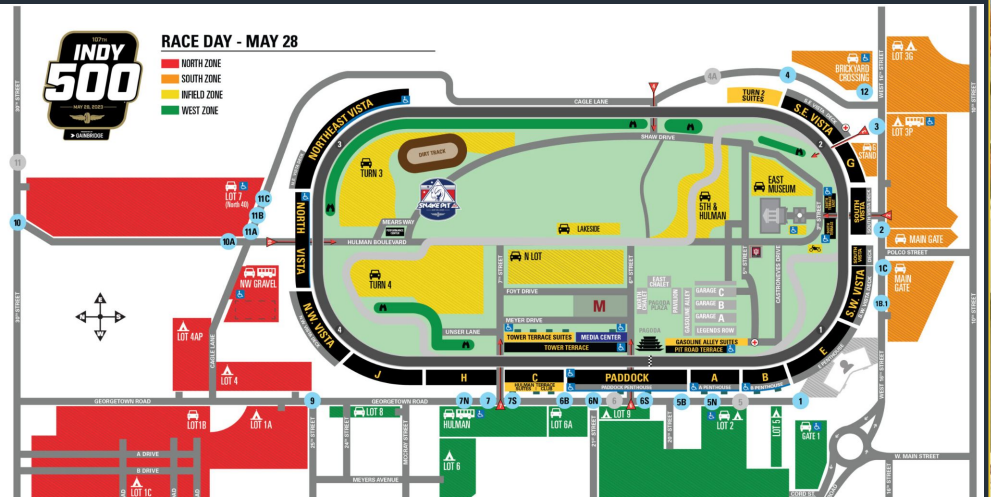





WT
WHITING-TURNER
WT
WHITING-TURNER

# Shell Eco-Marathon

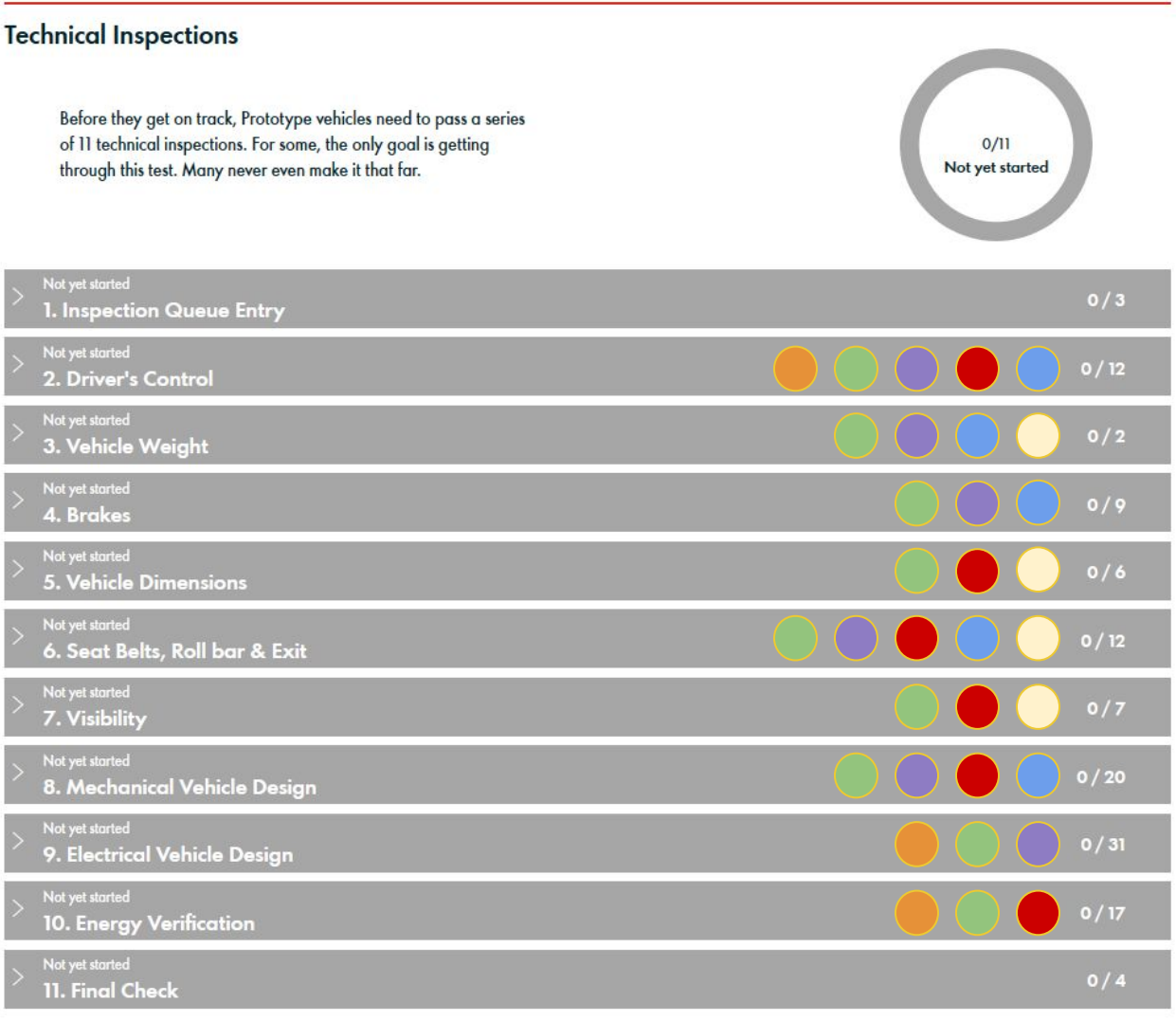
- Worldwide Engineering Competition
  - WAY less fast and money intensive version of Formula 1
- Multiple Categories: Prototype and Urban
- Different Powertrain Systems: Hydrogen, ICE-Hybrids, & Battery EVs












# Why Did We Do This As an E90?

- Structural Design
- Dynamics & Mechanical Design
- Fluid Mechanics
- Vibration Response
- Safety Systems
- Electrical Systems & Hydrogen Configuration

# Prior Vehicles and Past Troubles


Current Status


2018 Front Suspension


2012 - Original Chassis Construction



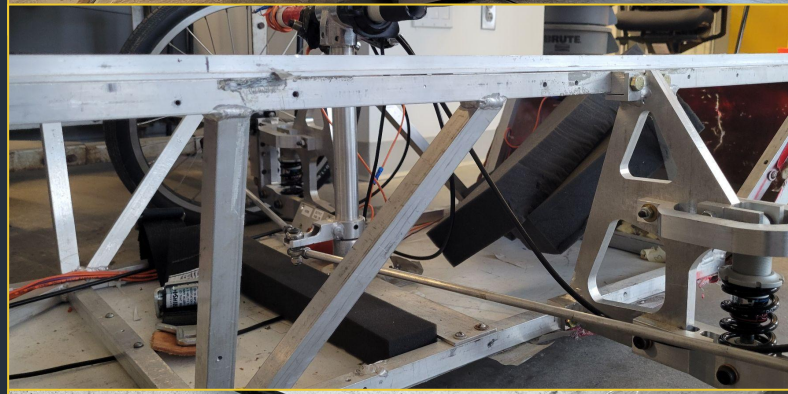
2015 Competition Team

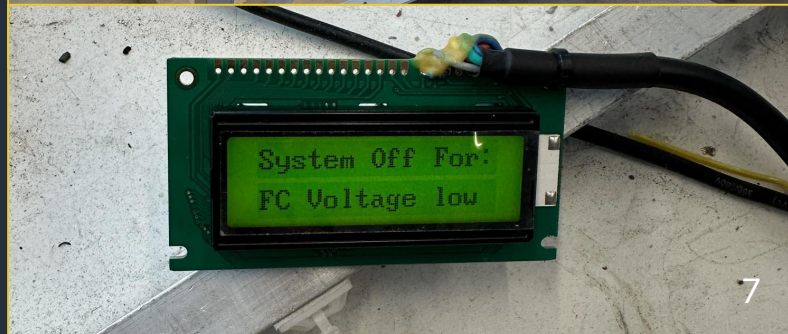
# 2018 and 2024 are different times

Current Issues

- Outdated Chassis Model & Materials
- Hydraulic Brakes
- Outdated Steering
- New front and rear suspension
- Driver Safety Systems
- Fuel Cell Woes







FROM CURRENT





Work cut out for us?





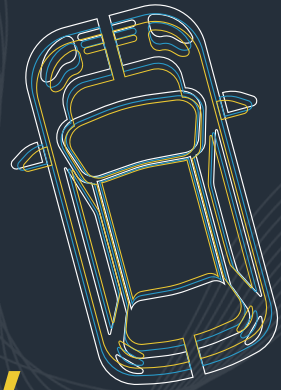
To New Standard

# 01

# Where Do We Start?

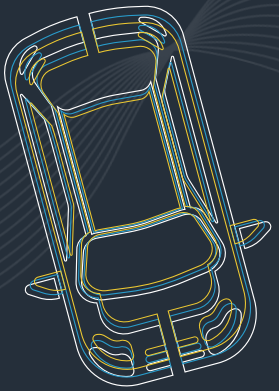
# Fuel Efficiency is King!

Optimize the kilograms of hydrogen used to drive **3.9kms in less than 35 mins**.



# Hypothesis: Weight #1 Priority

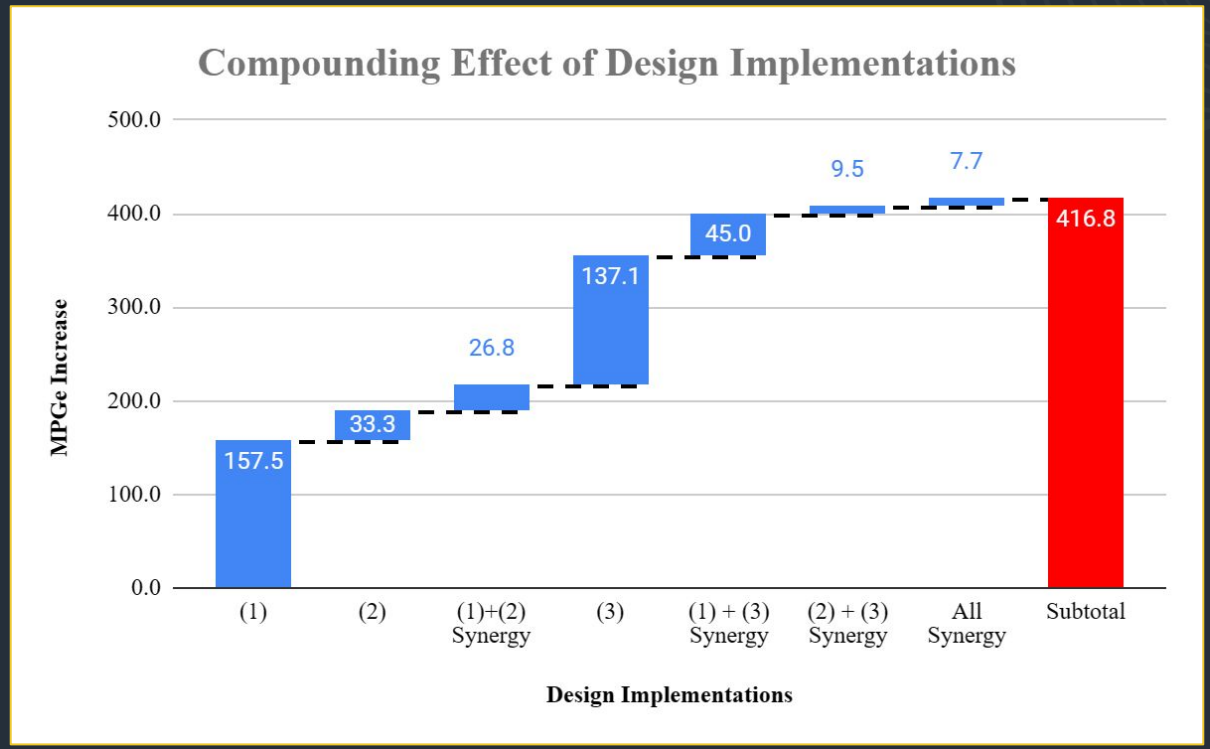
First efficiency model predicted an increase of **2-3 MPGe/lb saved.**



# Lots of opptys in new design

Material swaps, joint designs, fuel cell compactness, suspension redesign, aerodynamics, etc.

# Designing the Best Implementation Order

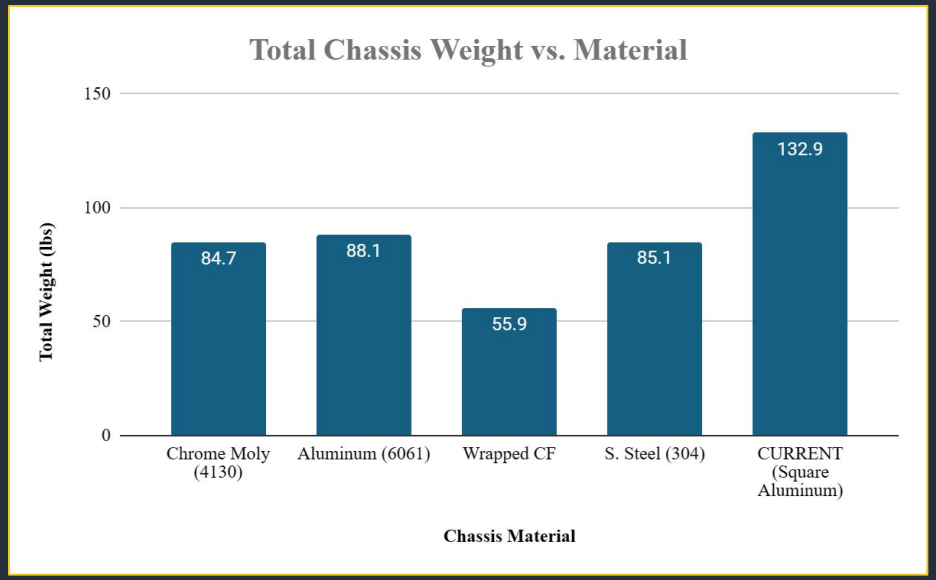


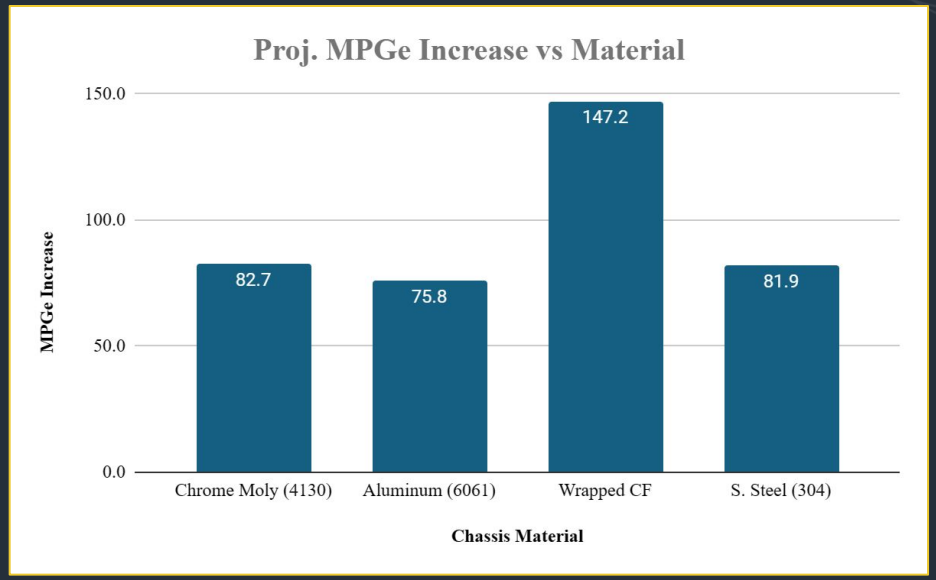

1. Chassis Weight Reduction by 33%

2. Frontal Area Reduction by 25%

3. Thermal Efficiency Increased by 10%

# What material makes the most sense?



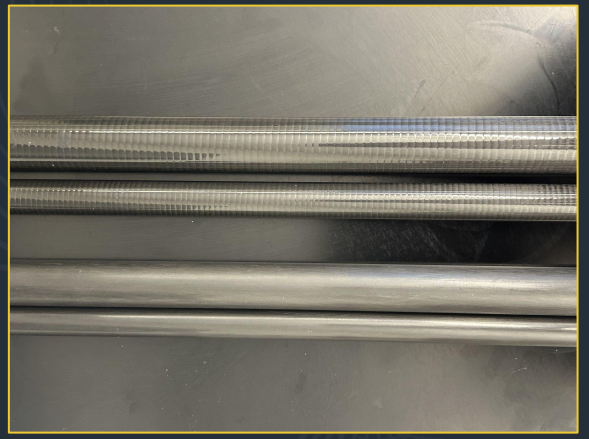



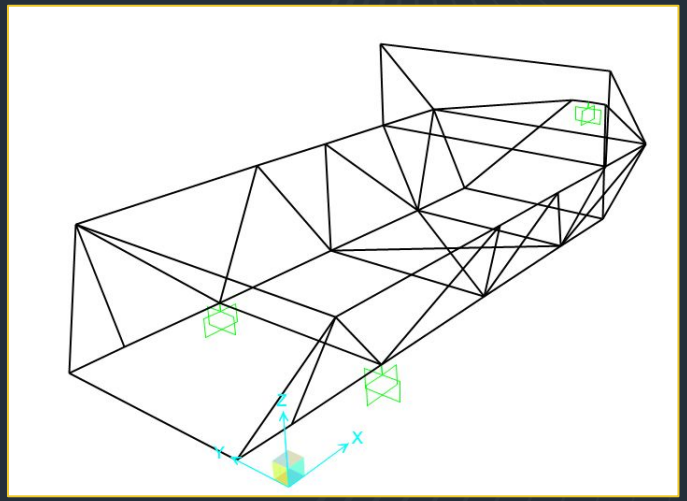

Wrapped CF can reduce chassis by 66%.....

...and increase overall efficiency by over 25%!!

# 02:

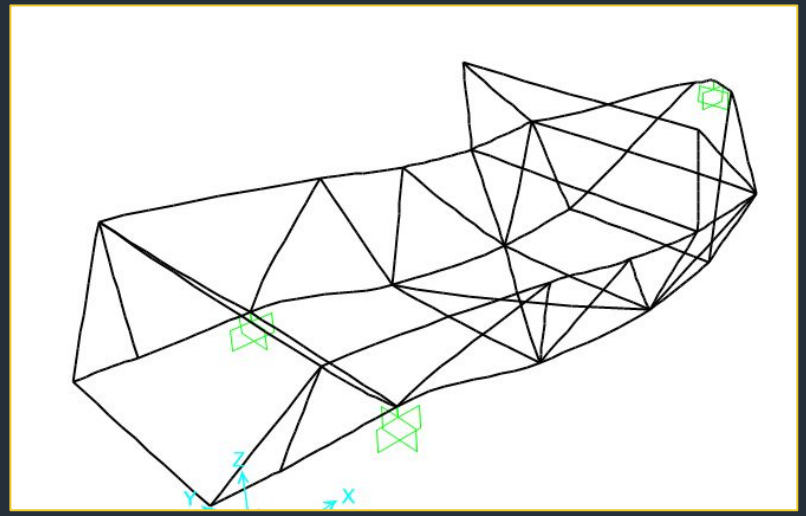
## New Chassis & Materials Research



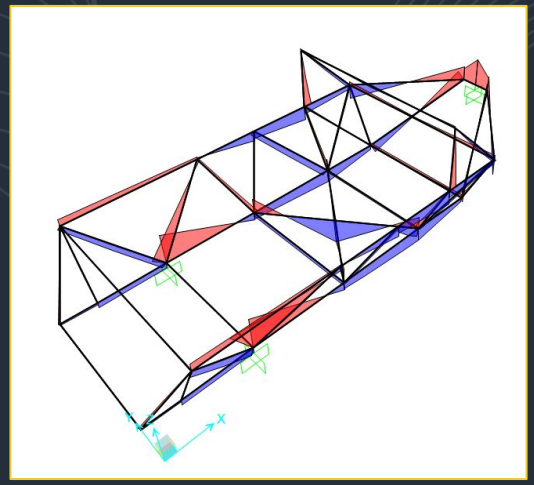


# SAP Model

Displacement

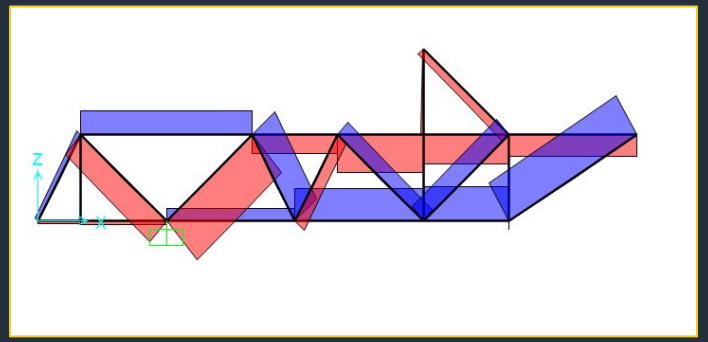



Maximum Moments




+ ~500 lb-in
- ~600 lb-in

Maximum Forces




T ~ 90 lbs
C ~ 105 lbs

# Connection Methods: Carbon Fiber Tubes

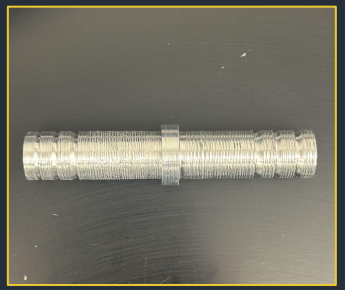
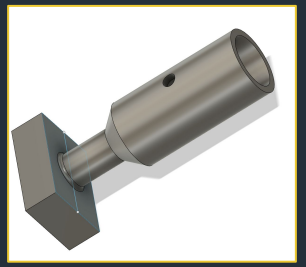
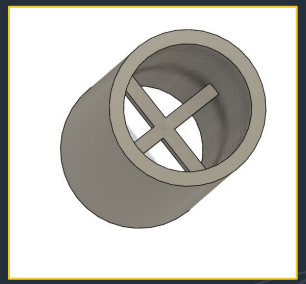
Research Credit: Rządkowski, et. al (2021). Evaluation of Bonding Gap Control Methods for an Epoxy Adhesive Joint of Carbon Fiber Tubes and Aluminum Alloy Inserts. Warsaw University of Technology




+

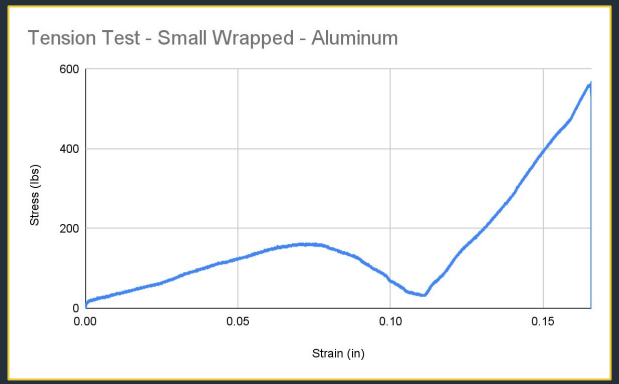


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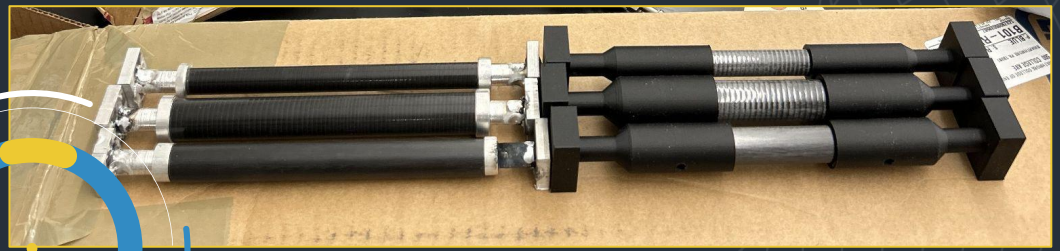
| | Tension Tests | Bending Tests | Conclusions: |
|---|---|---|---|
| Aluminum: |  |  | Great in tension, compression, and bending |
| 3D Prints: |  |  | Fasteners needed -> Great in axial tension and compression |

# Tension Tests





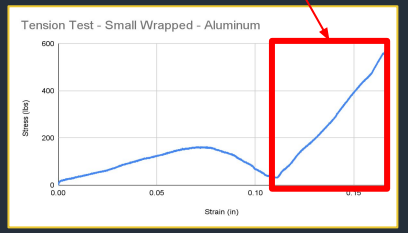


# Tension Test Results

**Maximum Simulated Tensile Force (200 lb Driver): 90 lb**
**Design Goal: 300 lb**

Linear Elastic Region




**Our Results**

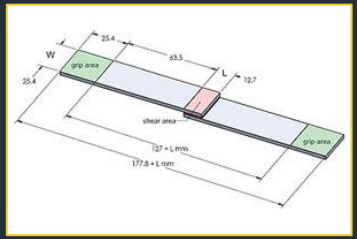
**Table X. Overview of Performance of Epoxy in Tension Test.**

| Material/Type | Fastener? | Mode of Failure | Ultimate Strength | Maximum Displacement* | Shear Modulus** |
|---|---|---|---|---|---|
| SW/3D | No | Epoxy | 222 lbs | .051 | 1184 lb/in^2 |
| LW/3D | No | Epoxy | 129 lbs | .04 | 583 lb/in^2 |
| SW/3D | Yes | Test Sample Failure | 383 lbs | .25 in | - |
| LP/Al | No | Epoxy | 422 lbs | .06 in* | 2141 lb/in^2 |
| SW/Al | No | Epoxy | 561 lbs | .05 in* | 3653 lb/in^2 |
| LW/Al | No | Epoxy | 598 lbs | .055 in* | 2327 lb/in^2 |

*Aluminum samples had welds crushed in beginning phases - displacement is estimated from the last major drop.

**Modulus of Elasticity was estimated using the final linear region before plastic deformation and failure. The estimated thickness of the epoxy is .01 in for all samples. The area of interaction is 2" long on two sides for the relevant diameter for the epoxy.

**3M Data Sheets: Overlap Shear Strength (ASTM D1002)**




**ABS/Polycarbonate: 450 lb/in^2**

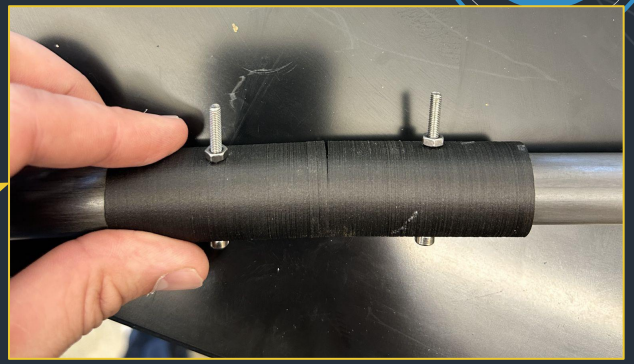
**Aluminum: 2500 lb/in^2**

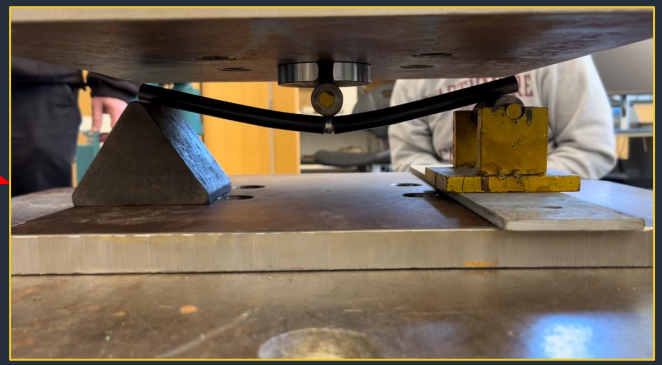
# Bending Results - On Joint

**Maximum Simulated Moment (200 lb Driver): 600 lb-in**
**Design Goal: 1800 lb-in (C)**

**Table X. Max Loads Before Failure - Three Point Bending Test**

| Tube | Joint | Failure Mechanism | Max Load | Max Moment |
|---|---|---|---|---|
| LW | 3D Collar | Break along print layer | 151 lbs | 434 lb-in |
| LP | 3D Collar | Break along print layer | 98 lbs | 281lb-in |
| LW | 3D Insert | Sheer through print core | 96 lbs | 276 lb-in |
| SW | Aluminum Insert | Aluminum bent inside of carbon fiber | 342 lbs | 983 lb-in |
| LW | Aluminum Insert | Aluminum breaks through side of tube | 1173 lbs | 3372 lb-in |
| LP | Aluminum Insert | Longitudinal crack in tube | 90 lbs | 258 lb-in |





# Costs of Weight Reduction & Efficiency Gain

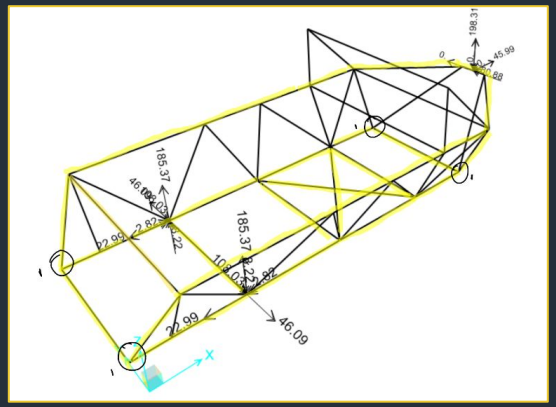
| CF Type | Connection Type | Current Wt Reduction % | Cost/lb saved ($) |
|---|---|---|---|
| SW | 3D collar | 30.8% | $30.20 |
| LW | 3D collar | 24.9% | $57.39 |
| SW | AL inserts | 9.9% | $76.98 |
| LW | AL inserts | 7.9% | $116.85 |

| CF Type | Connection Type | Proj MPGe Increases | Cost/ MPGe Gained ($) |
|---|---|---|---|
| SW | 3D collar | 229.4 | $9.86 |
| **LW** | **3D collar** | **175.2** | **$19.86** |
| SW | AL inserts | 60.7 | $30.52 |
| **LW** | **AL inserts** | **47.4** | **$47.14** |

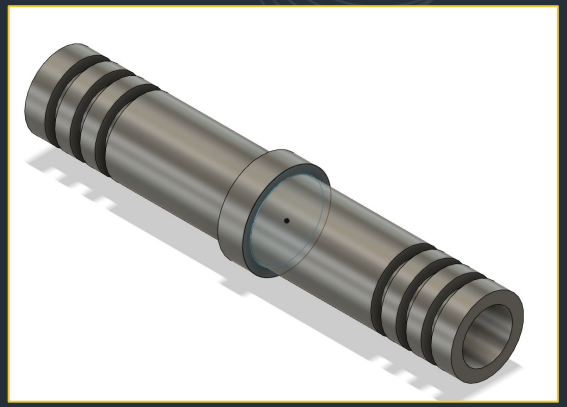
**However, 3D print collar methods were more cost effective in their weight reduction...**

**...and they more than tripled the expected efficiency gains if the entire chassis used them!**

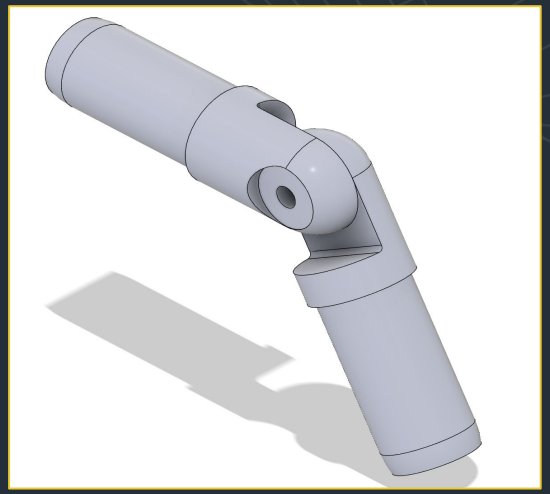
# A Combination of Inserts & Collars




Members with Aluminum joinery

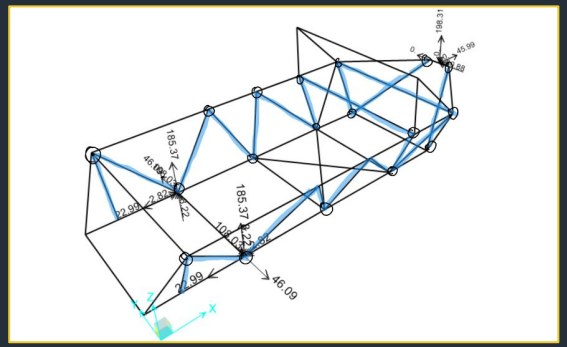


Machined Aluminum Hollow Insert

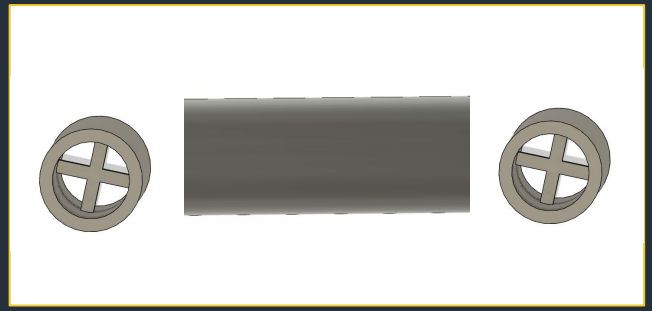


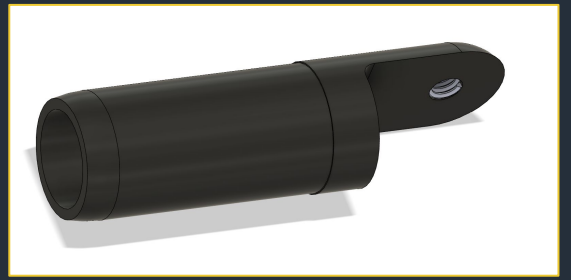
Clevis Threaded Connectors

Members with 3D print joinery




Carbon fiber+fiberglass Collar





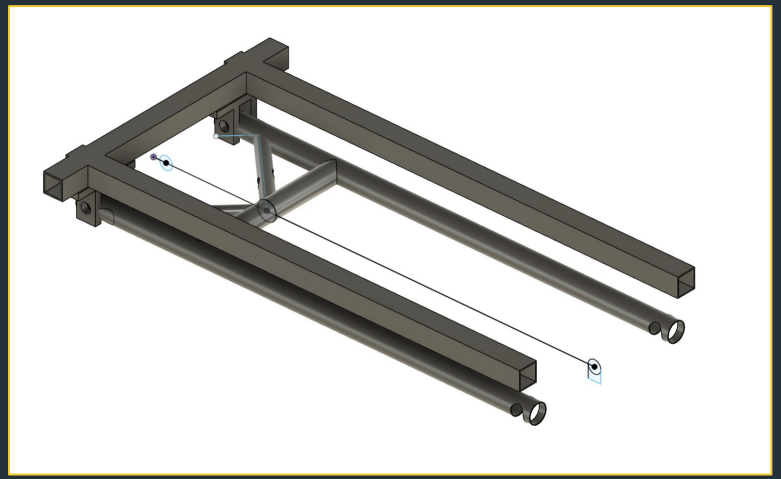
# 03:

# Additional Mechanical Systems





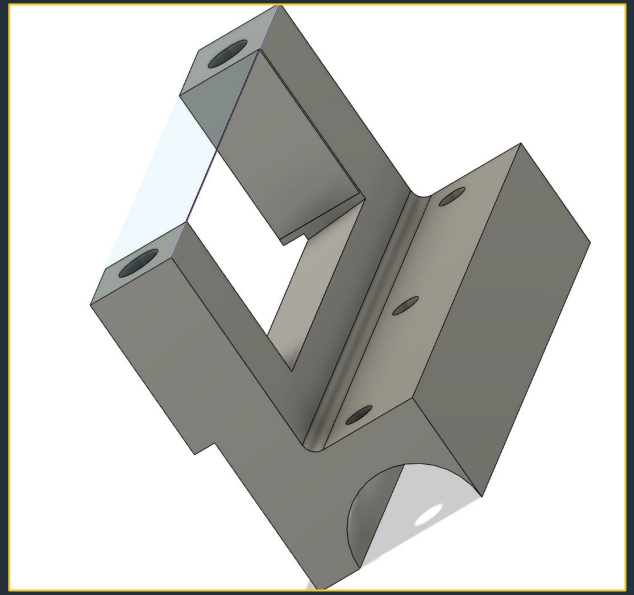

# Suspension Design, Fabrication, and Install



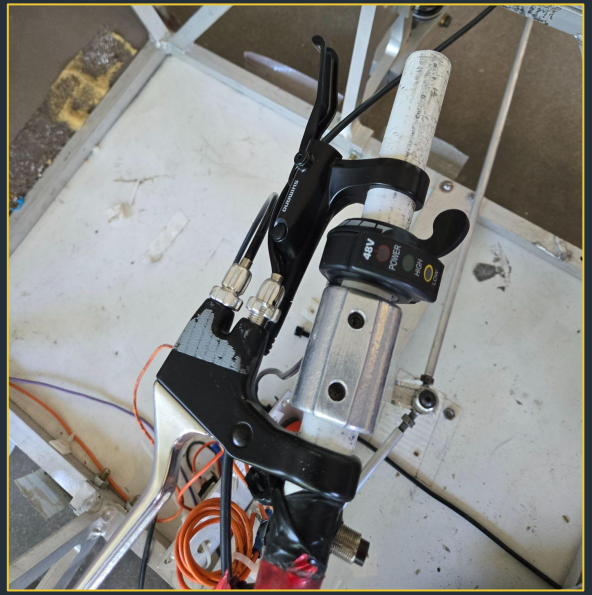




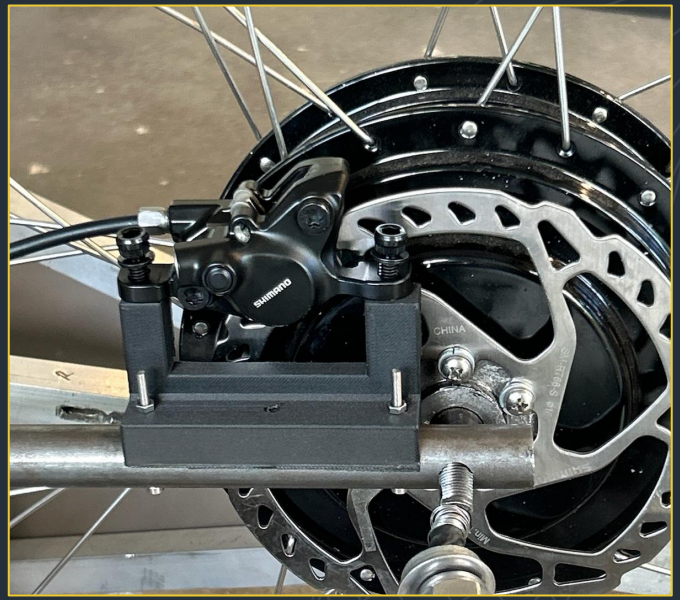
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**Step 1:** Brake Attachment on suspension is needed




**Step 2:** Remove the old rear brakes and wire new hoses and lever

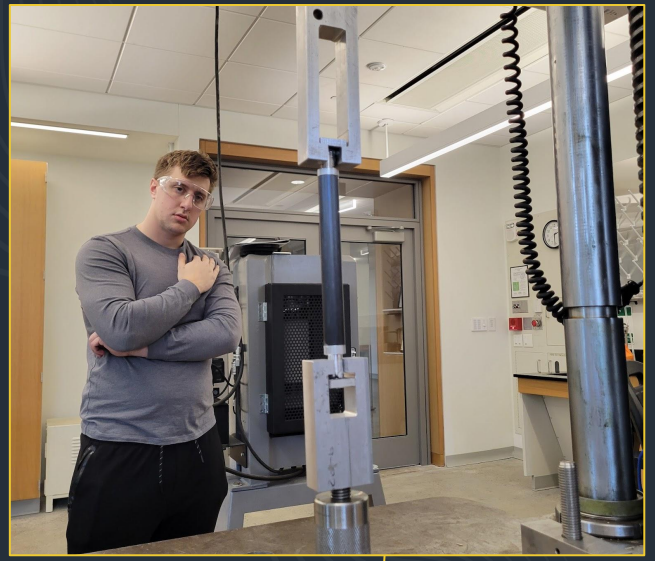



**Step 3**: Attach the new brake calipers on new disc rotor

# A Crash Course on Hydraulic Disc Brakes

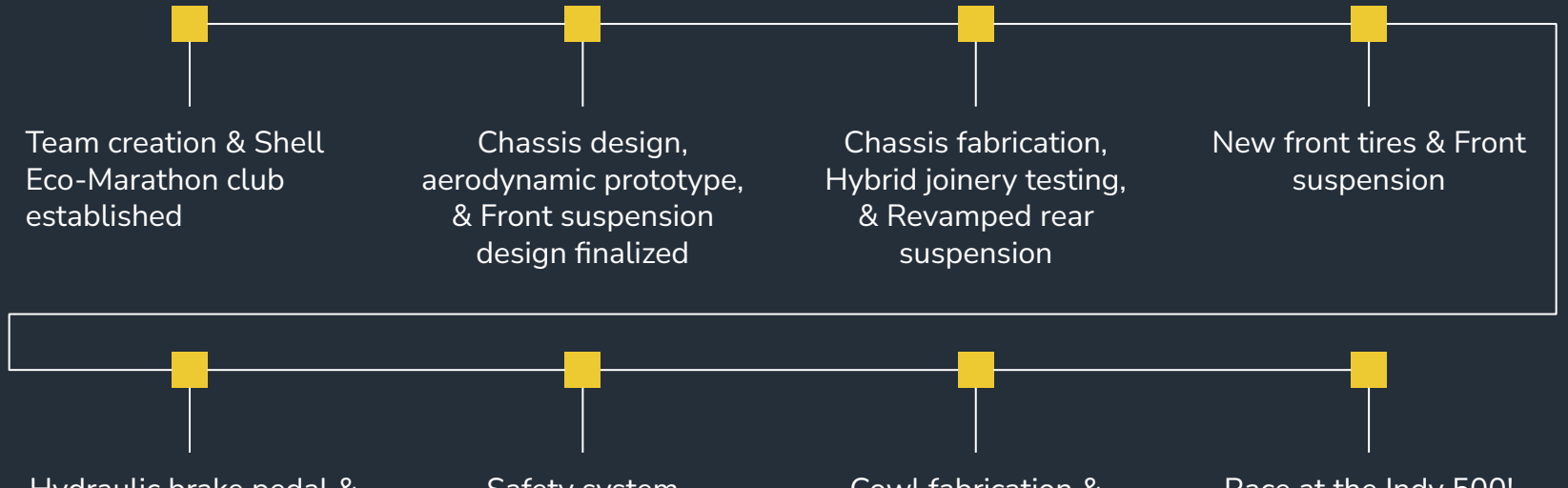
# 04:

# Conclusions & Future Considerations





# The Playbook for the Future



# Acknowledgments & Reflections







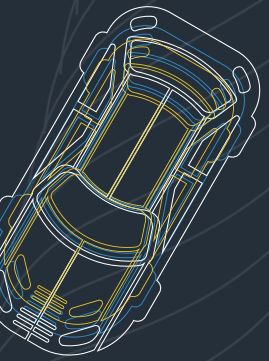


# Thanks!

Do you have any questions?

jchai1@swarthmore.edu
bhorvat1@swarthmore.edu

Vehicle Demo during the Break!



# Benchmark Tests (Speeds)



Inside Benchmark Speed Tests with a power error margin of 0.55%

| Test Trial | Max Velocity (m/s) | Max Acceleration (m/s^2) | Estimated Power Rating (J/s) |
|---|---|---|---|
| 1 | 6.25 | 1.28 | 408.4312005 |
| 2 | 6.62 | 4.04 | 1103.399375 |
| 3 | 6.75 | 3 | 867.3426618 |
| 5 | 6.5 | 6.2 | 1598.847531 |
| **Average** | **6.53** | **3.63** | **994.505192** |

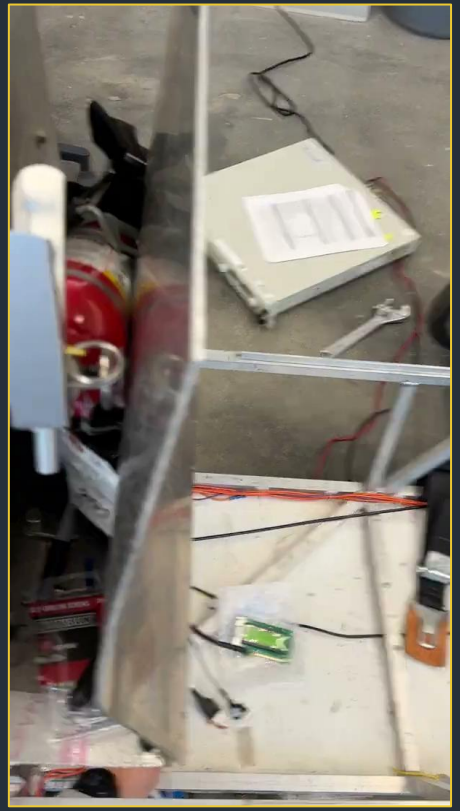
Outside Benchmark Speed Tests with a power error margin of 36,48%

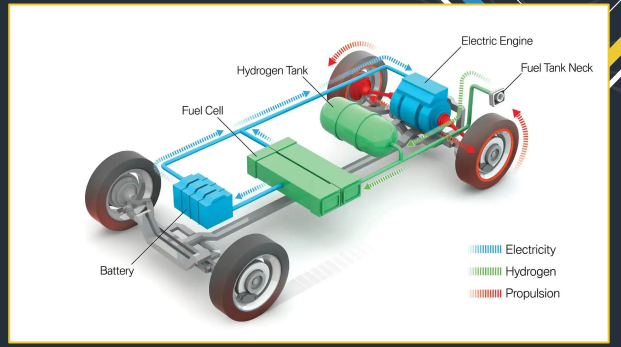
| Test Trial | Max Velocity (m/s) | Max Acceleration (m/s^2) | Estimated Power Rating (J/s) |
|---|---|---|---|
| 1 | 9.205 | 0 | 168.9712 |
| 2 | 10.6 | 0.1 | 233.4941515 |
| 3 | 8.9 | 4.1 | 1503.026818 |
| **Average** | **9.57** | **1.4** | **635.16** |

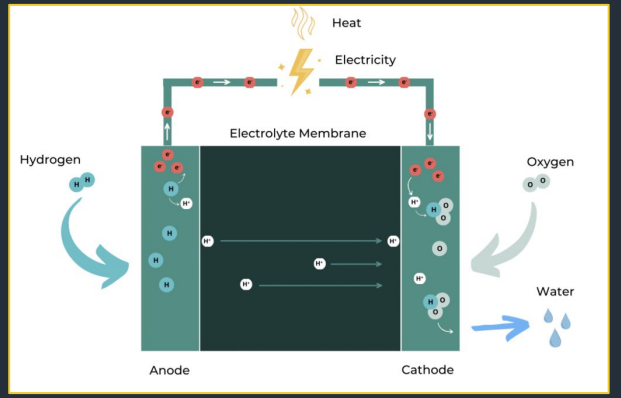
# Fuel Cell Renewal & Feasibility Assessment

## How does a HFCV work?

- $H_2$ gas moves along a membrane where it merges with $O_2$ gas to make water and electricity
- An electric motor powers the vehicle -> 0 emission kinematic propulsion!
- Solves the issues of current EVs where the weight is the main deterrent





# Efficiency and Kinematics Simulation

MODEL 2

| Control Panel | |
|---|---|
| % of hydrogen leaked | 0 |
| Fuel Cell Thermo-Efficiency | 35% |
| Distance Traveled (m) | 15330 |
| Average Speed (kph) | 30 |
| Density of hydrogen (kg/m^3) | 0.0838 |
| Asphalt Kinetic Friction Coefficient | 0.5000 |
| Estimated Drag Coefficient | 0.6500 |
| Esimtated Frontal Area (m^2) | 0.5000 |
| Density of Unlead. Gas (kg/L) | 0.765 |
| Net Caloric Value (H2) (kJ/kg) | 119,930 |
| Maximum power of HFC (J/s, W) | 1,000 |
| Density of air (kg/m^3) | 1 |
| Net Caloric Value (Unlead. Gas) (kJ/kg) | 42,900 |
| **Total Mass of Wrapped CF (lbs)** | 162.0879333 |
| **Total Mass of Current (lbs)** | 243.13 |

| Energy Used Calculations | CF | Current |
|---|---|---|
| Mass (kg) | 73.52 | 110.28 |
| KE at avg vel. (J) | 2552.84 | 3829.26 |
| Time taken until avg kph (secs) | 2.55 | 3.83 |
| Acceleration of vehicle (m/s^2) | 1.63 | 1.09 |
| Distance accelerated(m) | 5.32 | 7.98 |
| Distance at avg vel (m) | 15324.68 | 15322.02 |
| Drag Force Losses (J/s) | 121.69 | 121.69 |
| Friction losses (J/s) | 229.76 | 344.63 |
| Time to complete track (sec) | 1841.51 | 1842.47 |
| Total Frictional energy lost (J) | 647186.53 | 859182.09 |
| Total energy used(J) | 649739.37 | 863011.35 |

| Fuel Efficiency Calculations | CF | Current |
|---|---|---|
| Required energy from fuel cell (kJ) | 1856.398206 | 2465.746716 |
| **Hydrogen Needed (m^3)** | **0.1848019875** | **0.2454618263** |
| Gasoline Equivalency (kJ/m^3 of gas) | 32801340 | 32801340 |
| Amount of equivalent unlead. Gas consumed (m^3) | 0.000056595194 | 0.00007517213 |
| Fuel Efficiency (mi/m^3) (IN GASOLINE) | 168311.4189 | 126717.4012 |
| **Fuel Efficiency (MPGe)** | **637.1281546** | **479.677639** |
| **Fuel Efficiency (km/m^3) (H2)** | **82.95365328** | **62.45370302** |
| **Fuel Efficiency (mi/m^3) (of H2)** | **51.54499449** | **38.8069199** |

| | |
|---|---|
| **Increased Efficiency (mi/m^3) (H2)** | **12.73807459** |
| **Increased Efficiency (mpge)** | **157.4505156** |